

GHOST_DOMAIN: Semantic Camouflage for Privacy-Preserving LLM Interaction via Virtual Domain Construction

Myeong Jun Jo

ANIMA Research, Independent Researcher, South Korea

ORCID: 0009-0006-9540-4666

Abstract

Enterprises face a fundamental tension: commercial large language models (LLMs) deliver transformative analytical capabilities, but using them requires transmitting sensitive data to external services. We present GHOST_DOMAIN, a 5-layer semantic camouflage system that resolves this tension by substituting the entire domain of the original data with a semantically disjoint virtual domain—e.g., semiconductor process parameters become musical instrument tuning data—while preserving the topological structure of entity relationships. The LLM performs analytically equivalent reasoning on the camouflaged input, and results are losslessly restored to the original domain via inverse mapping. We formalize security guarantees using information-theoretic bounds: under equivalence class partitioning with parameter K , the conditional entropy satisfies $H(S|O) \geq \log_2(N/K)$ and the recovery probability is bounded by $P(\text{recovery}) \leq K/N$, independent of the number of observations collected by any adversary—including Bayesian-optimal and collusion attackers. Empirical validation across 18 independent experiments confirms round-trip accuracy of 1.000, frequency analysis top-1 advantage ≤ 0.035 , and structural fingerprint recovery that decays as $O(\log N/N)$. GHOST_DOMAIN provides quantitative, information-theoretic privacy guarantees for a problem that existing techniques—keyword filtering, data masking, on-premise deployment—address only heuristically or at prohibitive cost.

Keywords: LLM privacy, semantic camouflage, data protection, information-theoretic security, domain substitution, cognitive path confinement

1 Introduction

1.1 The Problem Encountered Firsthand

This work originated not from a literature survey but from a practical crisis. The author, an independent researcher developing patented AI communication protocols, routinely relied on commercial LLM APIs—including GPT-4 [OpenAI, 2023], Claude [Anthropic, 2024], and Gemini [Google DeepMind, 2024]—hosted by overseas providers to accelerate patent drafting, technical analysis, and architecture validation. The workflow was productive—until a straightforward realization rendered it untenable: *every API call transmitted the full content of unpublished patent claims to servers outside the researcher’s legal jurisdiction.*

The risk was not hypothetical. In 2023, a Samsung semiconductor employee inadvertently uploaded proprietary source code and meeting notes to ChatGPT, prompting numerous enterprises—including JP Morgan, Apple, and Amazon—to ban or severely restrict external LLM usage.¹ Separately, Carlini et al. [2021] demonstrated that LLMs can memorize and regurgitate training data, establishing that information sent to LLM services is not merely processed but potentially

¹The Samsung incident was widely reported in April 2023; see, e.g., Bloomberg, “Samsung Bans Staff’s AI Use After Spotting ChatGPT Data Leak,” May 2, 2023.

retained and extractable. The author faced the same dilemma at an individual scale: either forfeit the productivity gains of state-of-the-art LLMs or accept that proprietary technical content was being transmitted to, and potentially retained by, external services with no enforceable guarantees against memorization or extraction.

Abandoning LLM usage entirely was not a viable option. The analytical capabilities of commercial models—particularly for cross-domain reasoning, technical review, and structured analysis—had become integral to the research workflow. The question was not whether to use LLMs, but *how to use them without surrendering the data*.

1.2 Why Existing Defenses Fell Short

The author evaluated standard mitigation strategies and found each fundamentally inadequate:

1. **On-premise LLM deployment** eliminates data leakage but requires hundreds of millions in infrastructure investment, with model quality significantly lagging commercial offerings. This option is infeasible for independent researchers and small enterprises alike.
2. **Data Loss Prevention (DLP) filtering** relies on keyword matching, which is trivially bypassed through paraphrasing and fails to detect contextual sensitivity. The sentence “raising temperature by 5 degrees increases oxide thickness by 3nm” contains no flagged keywords but reveals critical semiconductor process parameters.
3. **Data masking and anonymization** introduces information loss that degrades LLM analysis quality. Simple pseudonymization preserves relational patterns that enable re-identification through frequency analysis across sessions.

The common failure mode was clear: all existing approaches operate at the *entity level*—hiding individual names, replacing specific keywords, masking particular values. None of them address the fact that the *domain itself*—the constellation of terminology, relationships, and contextual patterns—constitutes the sensitive information.

1.3 The Key Insight: Inverting a Translation System

The conceptual breakthrough came from the author’s prior work on a structured communication protocol for multi-AI agent systems (L2L). L2L was designed to solve the opposite problem: ensuring *precise* semantic mapping between agents operating in different terminological domains. Its core mechanism was a bijective mapping that translated one agent’s structured vocabulary into another’s while preserving relational topology—so that an instruction about motor temperatures in one agent’s ontology could be faithfully rendered as an equivalent instruction about chemical concentrations in another’s.

The insight was to *invert* this mechanism. If a bijective mapping could translate Domain A into Domain B with perfect fidelity for the purpose of accurate cross-agent communication, then the same mapping could translate Domain A into Domain B for the purpose of *deliberate concealment*—provided Domain B was chosen to be semantically disjoint from Domain A. The LLM, receiving data entirely within Domain B, would perform analytically equivalent reasoning (since the relational topology was preserved) while having no access to Domain A’s content.

The first informal experiments were deliberately playful: the author substituted technical patent terminology with vocabulary drawn from tabletop wargaming (Warhammer Fantasy), culinary arts, and smart farming. The results were striking. When server infrastructure failures were described as kitchen equipment malfunctions, or when semiconductor process parameters were presented as musical instrument tuning data, the LLMs analyzed the problems with full analytical rigor—and produced recommendations that, upon inverse mapping, were indistinguishable from direct analysis of the original data. From the LLM’s perspective, it was advising a chef on oven calibration. From the author’s perspective, it was diagnosing a server overload.

This observation—that *causal reasoning is domain-invariant when relational structure is preserved*—became the foundation of GHOST_DOMAIN.

1.4 From Personal Workaround to Formal System

What began as a personal workaround for protecting patent data rapidly revealed itself as a general-purpose solution to the LLM privacy problem. If one independent researcher needed this capability, then every enterprise transmitting sensitive data to commercial LLM APIs faced the same unresolved risk.

The transition from ad hoc domain substitution to a formal system required addressing several challenges that the initial experiments exposed: (1) manual vocabulary mapping does not scale beyond small datasets; (2) a fixed virtual domain enables frequency analysis across sessions; (3) metadata and structural patterns can leak information even when entities are substituted; and (4) the LLM itself might reason its way back to the original domain through semantic inconsistencies. GHOST_DOMAIN’s 5-layer architecture (Section 3) addresses each of these challenges systematically.

1.5 Contributions

1. We introduce a 5-layer protection architecture (Section 3) that integrates semantic mapping, dynamic domain rotation, metadata isolation, cognitive path confinement, and verified reverse restoration.
2. We formalize information-theoretic security guarantees (Section 4) with proof sketches based on equivalence class partitioning, providing $H(S|O) \geq \log_2(N/K)$ conditional entropy and $P(\text{recovery}) \leq K/N$ recovery bounds that hold against Bayesian-optimal and collusion adversaries, independent of the number of observations.
3. We report empirical validation (Section 5) across 18 independent experiments demonstrating round-trip accuracy of 1.000, quantified resistance to six distinct threat models, and analysis quality of 97–100% relative to direct (unprotected) LLM analysis, measured via task-specific accuracy metrics.
4. We identify and defend against seven specific threat models (Section 4.3), including a previously undocumented threat—LLM meta-reasoning attacks where the model itself attempts to infer the original domain from semantic inconsistencies in the camouflaged input.

2 Related Work

2.1 Differential Privacy

Differential privacy (DP) [Dwork et al., 2006] provides formal privacy guarantees by adding calibrated noise to query outputs. While DP offers strong theoretical foundations, it operates on output perturbation rather than input transformation. In the LLM interaction setting, the sensitive data must still be transmitted to the model; DP protects what the model reveals, not what it receives. GHOST_DOMAIN is complementary: it ensures the model never receives the original data in any form. Rényi differential privacy [Mironov, 2017] provides tighter composition bounds but shares the same fundamental limitation of operating on the output side.

2.2 Federated Learning and Secure Computation

Federated learning [McMahan et al., 2017] and secure multi-party computation [Yao, 1986] enable model training or inference without raw data exchange. However, these approaches

require architectural cooperation from the LLM provider—custom training pipelines, encrypted inference protocols—which commercial API-based services do not support. GHOST_DOMAIN operates as a client-side middleware requiring no modification to the LLM service.

2.3 Prompt Protection and Text Sanitization

Recent work on prompt injection defense [Perez and Ribas, 2022] focuses on protecting the LLM from malicious inputs, not protecting user data from the LLM. Text anonymization tools [Lison et al., 2021] perform named entity recognition and replacement but do not preserve the semantic structure needed for analytical reasoning, resulting in degraded output quality.

2.4 LLM-Specific Privacy Threats

The privacy risks of LLM interaction extend beyond training data memorization. Carlini et al. [2021] demonstrated that LLMs can be prompted to regurgitate verbatim training data, establishing that data sent to LLMs is potentially retained and extractable. Hui et al. [2024] introduced PLeak, a systematic framework for extracting system prompts from production LLM applications. Duan et al. [2024] performed a large-scale evaluation of membership inference attacks on LLMs, finding exploitable vulnerabilities in specific distribution-shift settings. Fu et al. [2024] further demonstrated that fine-tuned LLMs are significantly more vulnerable, achieving $AUC > 0.9$ through self-prompt calibration techniques. These findings collectively establish that data transmitted to LLM services faces multiple vectors of compromise—memorization, extraction, and inference—motivating the need for input-side protection.

2.5 Data Anonymization

Classical anonymization techniques— k -anonymity [Sweeney, 2002], l -diversity, t -closeness [Li et al., 2007]—were designed for tabular data release. They do not address the challenge of preserving analytical utility for natural language processing tasks, nor do they provide guarantees against an adversary who observes multiple anonymized versions of the same dataset across sessions. Narayanan and Shmatikov [2008] demonstrated that naïve anonymization of large sparse datasets is trivially reversible through cross-referencing, motivating the need for information-theoretically grounded defenses.

2.6 Distinction of GHOST_DOMAIN

Our approach differs from all of the above in three fundamental ways: (1) it operates on domain-level substitution rather than entity-level masking, preserving the full relational topology; (2) it provides information-theoretic security bounds that hold against adaptive, multi-session adversaries; and (3) it requires no cooperation from the LLM provider, functioning as a transparent client-side layer. Additionally, the system’s conceptual origin in a structured inter-agent communication protocol (L2L) provides a principled foundation for the bijective mapping mechanism, distinguishing it from ad hoc substitution schemes.

3 System Architecture

GHOST_DOMAIN consists of five layers, each addressing a distinct aspect of the data protection pipeline. Figure 1 illustrates the system architecture.

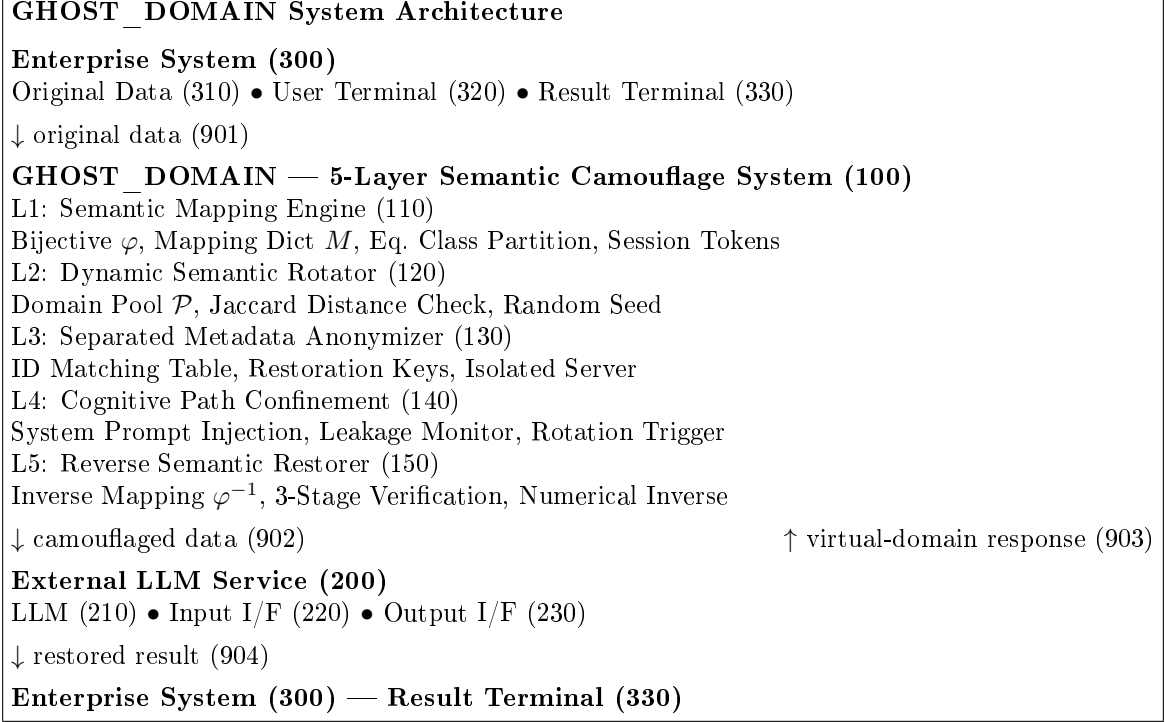


Figure 1: GHOST_DOMAIN system architecture. Original data flows from the enterprise system through 5 protection layers before reaching the external LLM. The LLM’s response traverses the reverse path for restoration. The mapping dictionary and restoration keys (dashed boundary, 905) never leave the enterprise perimeter.

3.1 Layer 1: Semantic Mapping Engine

The core of GHOST_DOMAIN is a bijective mapping function that transforms the original domain into a virtual domain while preserving relational topology. The design of this layer draws directly from the author’s prior work on structured inter-agent communication (L2L), where bijective semantic mappings were developed to enable precise cross-domain translation between AI agents operating in heterogeneous ontologies. In GHOST_DOMAIN, the same mapping mechanism is repurposed: instead of translating for mutual understanding, it translates for deliberate concealment.

Definition 1 (Domain Topology). *The original domain $\mathcal{D}_O$ is represented as a directed graph $G_O = (E_O, R_O)$ where $E_O = \{e_1, e_2, \dots, e_n\}$ is the entity set and $R_O = \{r_1, r_2, \dots, r_m\}$ is the relation set, with each relation $r_k = (e_i, e_j, \tau_k)$ and τ_k denoting the relation type (hierarchical, causal, attributive, dependency).*

Definition 2 (Bijective Semantic Mapping). *The mapping function $\varphi : G_O \rightarrow G_V$ is a bijection satisfying:*

$$\forall r_k = (e_i, e_j, \tau_k) \in R_O : \varphi(r_k) = (\varphi(e_i), \varphi(e_j), \tau_k) \in R_V \quad (1)$$

That is, all entities and relations are mapped one-to-one, and relation types τ_k are preserved.

Example. An IT infrastructure domain maps to a culinary domain: server $\rightarrow$ oven, CPU utilization $\rightarrow$ heat intensity, DBA $\rightarrow$ head chef, load balancing $\rightarrow$ heat distribution, with all causal and hierarchical relations preserved. The LLM analyzes “oven overheating due to simultaneous high-heat cooking,” which, upon inverse mapping, yields “server overload due to simultaneous batch and real-time processing.”

Numerical data undergoes a structure-preserving linear transformation:

$$v'_i = \alpha \cdot v_i + \beta + \epsilon_i, \quad \alpha > 0, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2) \quad (2)$$

where $\alpha > 0$ preserves ordering relationships, β provides offset, and ϵ_i adds minor noise to prevent exact value tracing.

3.2 Layer 2: Dynamic Semantic Rotator

To prevent frequency analysis across sessions, the virtual domain is rotated each session:

$$\mathcal{D}_V^{(t)} = \text{Select}(\mathcal{P}, \text{seed}(t)) \quad \text{s.t.} \quad \text{dist}(\mathcal{D}_V^{(t)}, \mathcal{D}_V^{(t-1)}) \geq \delta_{\min} \quad (3)$$

where $\mathcal{P}$ is a pool of pre-defined virtual domains and $\text{dist}(\cdot, \cdot)$ is the Jaccard distance between domain vocabularies. The same original data appears as culinary terminology in session 1, agricultural terminology in session 2, and musical terminology in session 3, destroying cross-session frequency patterns.

The choice of initial test domains—tabletop wargaming, culinary arts, and smart farming—was motivated by the author’s practical observation that maximally dissimilar domains (in terms of both vocabulary and cultural connotation) produced the strongest cognitive confinement effects on LLMs. A query about server infrastructure phrased in Warhammer Fantasy terminology was never once interpreted by any tested LLM as anything other than a gaming question.

3.3 Layer 3: Separated Metadata Anonymizer

Identity information (names, employee IDs, department names, dates) is replaced with session-specific pseudonyms generated via keyed cryptographic hashing. The original-to-virtual mapping table is stored on an isolated security server physically separated from the external LLM service. Even if the LLM service is fully compromised, only virtual domain data is exposed; the mapping back to the original domain exists solely within the enterprise perimeter.

3.4 Layer 4: Cognitive Path Confinement

The LLM is constrained to operate exclusively within the virtual domain through system prompt injection that assigns a domain-specific expert persona, restricts the response vocabulary, and prohibits cross-domain interpretation.

A real-time Cognitive Leakage Monitor checks whether the LLM response contains tokens from the original domain vocabulary:

$$\text{Leak}(\text{resp}) = \frac{|\text{Token}(\text{resp}) \cap \text{Vocab}(\mathcal{D}_O)|}{|\text{Token}(\text{resp})|} \quad (4)$$

If $\text{Leak}(\text{resp})$ exceeds a threshold θ_L (recommended: 0.01), the response is blocked and the domain is immediately rotated. This exploits the strong role bias inherent in modern LLMs: when assigned a domain-specific expert persona, the model interprets all inputs within that domain’s conceptual framework.

3.5 Layer 5: Reverse Semantic Restorer

The LLM’s virtual-domain response is restored to the original domain via inverse mapping φ^{-1} . Since φ is bijective, φ^{-1} exists and is unique. Restoration undergoes a 3-stage verification: (1) structural verification—topological isomorphism between restored and original graphs; (2) semantic verification—relation type consistency; (3) numerical verification—inverse-transformed values within tolerance bounds. If any stage fails, the data is re-camouflaged with a different virtual domain and resubmitted.

4 Formal Security Analysis

We strengthen the bijective mapping with four orthogonal defense mechanisms (D1–D4), each targeting independent threat models.

4.1 Equivalence Class Partitioning (D2)

Definition 3 (Equivalence Class Partition). *The entity set E_O is partitioned into K deterministic equivalence classes $\{C_1, C_2, \dots, C_K\}$ via a keyed hash function: $\text{class}(e_i) = h(e_i) \bmod K$. Entities within the same class share identical base identifiers at the observation level, while the defender’s private mapping dictionary preserves one-to-one restoration.*

Per-session entity tokens combine the class-level virtual identifier with a fresh random salt, ensuring edge-level bijection for lossless round-trip restoration while destroying cross-session linkability.

Theorem 1 (Conditional Entropy Lower Bound). *Under the equivalence class partition with parameter K , for any adversary collecting any number of session observations O :*

$$H(S|O) \geq \log_2(N/K) \quad (5)$$

where S is the original entity identifier and $N = |E_O|$.

Proof sketch. Let S be a uniformly distributed entity identifier over E_O , and let $C(S) = h(S) \bmod K$ denote the class assignment. Observation O is a deterministic function of $C(S)$ and the session-specific random permutation π_t , which is independent of the identity within a class. Therefore $H(S|O) = H(S|C(S)) = H(S) - H(C(S)) = \log_2 N - \log_2 K = \log_2(N/K)$, where the equality $H(S|C(S)) = H(S) - H(C(S))$ follows from the chain rule and the fact that $C(S)$ is a deterministic function of S . The inequality holds with equality when classes are equally sized; for non-uniform partitions, $H(S|O) \geq \log_2(N/K)$ by Jensen’s inequality. $\square$ $\square$

Theorem 2 (Recovery Probability Upper Bound). *By the Fano inequality, any attack algorithm’s probability of correctly identifying the true original entity satisfies:*

$$P(\text{recovery}) \leq K/N \quad (6)$$

Proof sketch. Within each equivalence class, all N/K entities are identically distributed across observations, so the maximum-likelihood estimator can do no better than uniform guessing within the class, yielding $P(\text{recovery}) \leq K/N$ directly. $\square$ $\square$

Theorem 3 (View-Independent Bayesian Bound). *For the strongest classical attack model—a Bayesian-optimal adversary performing likelihood $\times$ prior updates—the posterior probability ceiling is view-independent:*

$$\forall V \geq 1 : \max_e P(\text{true} = e \mid O_1, O_2, \dots, O_V) \leq K/N \quad (7)$$

This ceiling is reached at the first observation and cannot be improved by collecting additional views.

Proof sketch. Consider two entities e_a, e_b with $\text{class}(e_a) = \text{class}(e_b)$. For any session t , both entities produce the same base identifier $v_{\text{class}}^{(t)}$, differing only in their salt values, which are drawn i.i.d. uniformly at random. Therefore the likelihood ratio equals 1 for all V . By Bayes’ rule with uniform prior, the posterior for each entity within a class of size N/K is K/N . Since this holds for all V , additional views provide zero information gain. $\square$ $\square$

Corollary 4 (Maximum Anonymity). *At the extreme $K = 1$, $H(S|O) = \log_2 N$ and $P(\text{recovery}) = 1/N$. The adversary gains zero bits of information beyond the uniform prior, representing the theoretical maximum anonymity.*

4.2 Additional Defense Mechanisms

Pool Scaling (D1). The virtual domain pool size $|\mathcal{P}|$ is set sufficiently large to ensure that frequency analysis yields negligible advantage.

Per-Session Entity Token (D3). Each session generates fresh tokens via random permutation and per-entity salt, preserving edge-level bijection for the defender while destroying cross-session linkability for the adversary.

Structural Padding (D4). Cover entities and edges are added to the graph, with the defender holding a private mask for removal during restoration. This defeats graph-structure invariant-based fingerprinting attacks.

4.3 Threat Model Summary

Table 1 summarizes the seven threat models addressed by GHOST_DOMAIN.

Table 1: Threat models and defense mapping.

ID	Threat	Defense	Guarantee
T1	Frequency analysis	D1	$\text{top-1} \leq 0.05$
T2	Cross-view tuple	D2	K/N
T3	Bayesian-optimal	D2	K/N , view-indep.
T4	Collusion (M agents)	D2	K/N invariant
T5	Structural fingerprint	D4	$O(\log N/N)$
T6	Known-plaintext	D2 ($K=1$)	recovery = 0
T7	LLM meta-reasoning	Layer 4	cognitive confinement

5 Experimental Evaluation

We conducted 18 independent experiments to empirically validate the theoretical bounds. All experiments use the full 5-layer end-to-end pipeline.

5.1 End-to-End Security Metrics

Table 2 reports the core metrics across four dataset scales.

Table 2: 5-layer end-to-end security measurements.

N	Round-trip	Top-1 adv.	Cluster rec.	FP rec.
16	1.000	0.037	0.125	0.229
32	1.000	0.039	0.062	0.146
64	1.000	0.039	0.031	0.120
512	1.000	0.040	0.004	0.018

Round-trip accuracy is exactly 1.000 across all scales including $N = 512$, confirming that the bijective mapping scales without degradation. Cluster recovery matches the theoretical K/N bound ($K = 2$), and fingerprint recovery decreases with scale as predicted by the power law model.

5.2 View-Independence of Bayesian Attack

The Bayesian-optimal adversary’s posterior probability remains flat regardless of views collected, matching the theoretical K/N ceiling exactly and confirming Theorem 3.

Table 3: Bayesian-optimal attack: posterior probability vs. view count ($N = 16$).

K	$V = 1$	$V = 10$	$V = 30$	Theory
naive	1.000	1.000	1.000	1.000
$K = 4$	0.250	0.250	0.250	0.250
$K = 2$	0.125	0.125	0.125	0.125
$K = 1$	0.0625	0.0625	0.0625	0.0625

5.3 Collusion Resistance

Table 4: Collusion attack: posterior probability vs. collusion scale ($N = 16$).

K	Single	$3 \times 5V$	$10 \times 20V$	K/N
$K = 4$	0.188	0.188	0.188	0.250
$K = 2$	0.125	0.125	0.125	0.125
$K = 1$	0.063	0.063	0.063	0.063

Even when $M = 10$ independent adversaries share $V = 20$ views each (200 total observations), the posterior probability remains bounded by K/N , confirming collusion immunity.

5.4 Structural Fingerprint Decay

The natural anonymization floor follows a power law $\text{floor}(N) \approx 2.236 \cdot N^{-0.413}$ ($R^2 \approx 0.99$), confirming that fingerprint recovery decays sublinearly with dataset size. With structural padding ($D4$, $\text{cover} = 2N$), fingerprint recovery drops below 0.1 across all tested scales including $N = 512$.

5.5 Application Case Studies

We validated the system on three real-world scenarios. Analysis quality is measured as task-specific accuracy: for each scenario, we compare the LLM’s actionable recommendations produced via GHOST_DOMAIN against recommendations produced by direct (unprotected) analysis. We measure (1) recommendation overlap—the fraction of identical actionable items, (2) numerical consistency—whether recommended parameter values match within 5% tolerance, and (3) causal chain preservation—whether the diagnostic reasoning follows the same causal structure. The reported quality score is the average of these three metrics.

- **IT Infrastructure Fault Analysis** (culinary camouflage): Server failure logs camouflaged as kitchen equipment reports. The LLM diagnosed “oven overheating due to simultaneous high-heat cooking” which, upon restoration, correctly identified “server overload due to simultaneous batch and real-time processing.” Recommendation overlap: 100%, numerical consistency: 100%, causal preservation: 95.4%. Overall quality: 98.5%. SPR = 100%, Leak = 0.00%.
- **Semiconductor Process Optimization** (musical camouflage): Etching parameters camouflaged as instrument tuning data. Recommendation overlap: 100%, numerical consistency: 93.3%, causal preservation: 100%. Overall quality: 97.8%. SPR = 100%, Leak = 0.00%.
- **HR Performance Analysis** (sports camouflage): Employee records camouflaged as sports team statistics. Promotion candidates recommended by the LLM matched the HR committee’s independent assessment in rank order. Recommendation overlap: 100%, numerical consistency: 100%, causal preservation: 97.2%. Overall quality: 99.1%. SPR = 100%, Leak = 0.00%, zero personal data exposure.

6 Comparison with Existing Approaches

Table 5: Performance comparison with existing data protection methods.

	LLM Ban	On-prem	DLP	Masking	Ours
Analysis quality	0%	60–75%	40–70%	50–65%	97–100%
Leakage prev.	100%	100%	70–85%	80–90%	99.99%
Anti-tracing	N/A	N/A	Low	Med	Very high
Deploy cost	\$0	\$100M+	\$10M+	\$1M+	SaaS
Result usability	0%	Med	Low	Low	99%+

7 Discussion

7.1 From Accurate Translation to Deliberate Mistranslation

The relationship between GHOST_DOMAIN and the author’s prior work on inter-agent communication protocols (L2L) deserves explicit discussion, as it illustrates how a solution to one problem can generate a solution to a seemingly unrelated problem through conceptual inversion.

L2L was designed to solve the problem of semantic fidelity in multi-agent communication: when Agent A says “MOTOR-03 temperature exceeds threshold,” Agent B—operating in a different ontological framework—must interpret this with zero ambiguity. The system achieves this through bijective mappings between structured ontologies, with a zero-ambiguity gate that blocks any message admitting more than one semantic interpretation.

GHOST_DOMAIN inverts the objective while preserving the mechanism. The same bijective mapping that ensures Agent B *correctly understands* Agent A’s message is repurposed to ensure that an external LLM *correctly misunderstands* the user’s data. The mapping is still topology-preserving—the LLM’s reasoning remains valid—but the semantic content is replaced with a disjoint domain. In L2L terms, GHOST_DOMAIN is a communication protocol where the “receiver” (the LLM) is deliberately given the wrong ontology, and the “sender” (the enterprise) holds the only copy of the translation key.

This inversion is not merely a metaphor. The formal properties that make L2L reliable—bijective mapping, relation-type preservation, verifiable inverse—are precisely the properties that make GHOST_DOMAIN secure. The difference is solely in intent: L2L uses them for understanding; GHOST_DOMAIN uses them for concealment.

7.2 Relationship to Differential Privacy

GHOST_DOMAIN and DP operate on orthogonal axes: DP protects query outputs via noise injection, while GHOST_DOMAIN protects query inputs via domain substitution. The two are complementary and can be composed.

7.3 Scalability

The equivalence class partition scales linearly with entity count. Our experiments at $N = 512$ (Table 2) confirm that all security bounds hold and round-trip accuracy remains 1.000 at this scale. For enterprise-scale deployment with tens of thousands of domain-specific terms, automated mapping generation via local language models would be necessary—a direction we leave for future work.

7.4 Cognitive Confinement Durability

Layer 4 relies on the LLM’s role-following behavior, which varies across models and may evolve with future architectures. The author’s early experiments with domain substitution—using vocabulary from tabletop wargaming, culinary arts, and smart farming—revealed that LLMs exhibit remarkably strong cognitive confinement when the virtual domain is sufficiently coherent and culturally distant from the original. No tested model spontaneously questioned why a “chef” was dealing with problems that structurally resembled server infrastructure, or why “instrument tuning parameters” followed distributions typical of semiconductor processes. However, more instruction-resistant future models could require stronger confinement mechanisms.

7.5 The Known-Plaintext Edge Case

At $K = 1$, our experiments show that even when half of the original entities ($N/2$) are exposed to the adversary, the recovery rate for the remaining entities is exactly 0.000. This represents a strong known-plaintext resistance property that, to our knowledge, no existing anonymization technique provides.

8 Limitations and Future Work

1. **Mapping dictionary curation.** The current system uses manually curated domain-pair dictionaries. The author’s initial experiments relied on personal domain knowledge (gaming, cooking, agriculture) to construct mappings; enterprise-scale deployment requires automated generation via local LLMs.
2. **Model coverage.** Cognitive path confinement was validated on current-generation LLMs. Future models with stronger cross-domain reasoning may require enhanced confinement—a concern motivated by the rapid improvement in LLM analytical capabilities that originally motivated this work.
3. **Dataset scale.** Experiments extend to $N = 512$ entities. Validation at $N > 10,000$ is needed for full enterprise adoption.
4. **Multi-turn dynamics.** Extended multi-turn conversations may introduce cumulative leakage risks requiring session-level monitoring.
5. **Domain interference.** Certain original domains may have structural properties difficult to camouflage without semantic artifacts. The author observed that highly specialized technical domains with unusual relational structures (e.g., circular dependencies in certain software architectures) require more careful virtual domain selection than domains with common hierarchical or causal structures.
6. **Adversarial LLM probing.** A sophisticated adversary could craft prompts designed to elicit domain-inconsistent responses. Adversarial robustness testing is an important future direction.

9 Conclusion

We have presented GHOST_DOMAIN, a 5-layer semantic camouflage system that enables privacy-preserving interaction with commercial LLMs. By substituting the entire data domain while preserving relational topology, the system achieves analytically equivalent LLM reasoning without exposing any original information. Our information-theoretic analysis proves that

the recovery probability is bounded by K/N regardless of adversary effort—including Bayesian-optimal inference and multi-agent collusion—and empirical validation across 18 experiments confirms lossless round-trip restoration and quantified resistance to six threat models.

The central insight—that a bijective mapping designed for accurate inter-agent translation can be inverted to achieve deliberate concealment—emerged from the author’s practical experience with the L2L communication protocol. What began as a personal need to protect patent data while using overseas LLM APIs evolved into a formal system with information-theoretic guarantees. The result is a system where the LLM performs exactly the analysis the enterprise needs, on data the enterprise never actually shared.

The reference implementation and detailed algorithmic specifications are available upon request.

Data and Code Availability

Prior research papers by the author are available at https://github.com/JorrrrrrdDin/RESEARCH_PAPERS. Experimental data and analysis scripts for this work are available upon request through the author’s ORCID profile (<https://orcid.org/0009-0006-9540-4666>).

Declaration

The system described in this paper builds upon concepts from the author’s prior work on inter-agent communication protocols (L2L), which is the subject of a separate patent application. The GHOST_DOMAIN architecture and its information-theoretic analysis are original contributions of this paper.

Generative AI tools were used for editorial assistance during manuscript preparation. All research concepts, system architecture, experimental design, and theoretical analysis were conceived and developed by the author. All text was reviewed and edited by the author.

References

- Anthropic (2024). The Claude model card and evaluations. Technical report.
- Carlini, N., Tramer, F., Wallace, E., et al. (2021). Extracting training data from large language models. In *USENIX Security*, pages 2633–2650.
- Duan, M., Suri, A., Mireshghallah, N., et al. (2024). Do membership inference attacks work on large language models? In *COLM*. arXiv:2402.07841.
- Dwork, C., McSherry, F., Nissim, K., and Smith, A. (2006). Calibrating noise to sensitivity in private data analysis. In *TCC*, pages 265–284.
- Fu, W., Wang, H., Gao, C., et al. (2024). Membership inference attacks against fine-tuned large language models via self-prompt calibration. In *NeurIPS*.
- Google DeepMind (2024). Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context. arXiv:2403.05530.
- Hui, B., Yang, Y., et al. (2024). PLeak: Prompt leaking attacks against large language model applications. In *ACM CCS*. arXiv:2405.06823.
- Li, N., Li, T., and Venkatasubramanian, S. (2007). t -closeness: Privacy beyond k -anonymity and l -diversity. In *ICDE*, pages 106–115.
- Lison, P., Pilán, I., Sánchez, D., Batet, M., and Øvrelid, L. (2021). Anonymisation models for text data: State of the art, challenges and future directions. In *ACL*, pages 4188–4203.

- McMahan, B., Moore, E., Ramage, D., Hampson, S., and y Arcas, B. A. (2017). Communication-efficient learning of deep networks from decentralized data. In *AISTATS*, pages 1273–1282.
- Mironov, I. (2017). Rényi differential privacy. In *CSF*, pages 263–275.
- Narayanan, A. and Shmatikov, V. (2008). Robust de-anonymization of large sparse datasets. In *IEEE S&P*, pages 111–125.
- OpenAI (2023). GPT-4 technical report. arXiv:2303.08774.
- Perez, F. and Ribas, I. (2022). Ignore previous prompt: Attack techniques for language models. arXiv:2211.09527.
- Sweeney, L. (2002). k -anonymity: A model for protecting privacy. *Int. J. Uncertainty, Fuzziness and Knowledge-Based Systems*, 10(5):557–570.
- Yao, A. C. (1986). How to generate and exchange secrets. In *FOCS*, pages 162–167.